

Week 2: Physical Layer: Guided Media and Digital Signaling and Wireless Communication and Networking

For the lectures we are learning the Physical Layer and wireless communication, how bits are physically transmitted over different media and the fundamental limits that govern communication systems.

We move from entropy (abstract information from previous lecture) to transmission constraints like noise, bandwidth, and signal degradation.

- 1) Guided Transmission Media: Different media are optimized for different tradeoffs:

Medium	Distance	Bandwidth	Cost	Use Case
Twisted Pair	About 100m	GHz	low	Ethernet
Coaxial	About 500m	GHz	Med	Cable Internet
Optical Fiber	100+km	THz	High	Backbone

The best one depends on the case and the constraints.

- 2) Signal Attenuation & dB

Signals lose power over distance therefore instead of multiplying losses, we use decibels (dB) to turn multiplication into addition making calculation much simpler.

Link Budget (account for all power gains and losses in a communication system can the signal reach the destination) : $P_{rx}(dBm) = P_{tx}(dBm) - L_{cable}(dB) + G_{tx} + G_{rx}$

- 3) Why Fiber Dominates ? Fiber Dominates because it has the following (better compared to Coaxial and Twisted Pair) Extremely low loss (0.2 dB/km) , Massive bandwidth (>10 THz), Long distances (100+ km)
- 4) Shannon-Hartley Theorem: $C = B \log_2 (1 + S/N)$ bits/s

Where: C = max data rate, B = bandwidth, S/N = signal-to-noise ratio. Capacity grows linearly with bandwidth and Capacity grows logarithmically with SNR(signal to noise ratio)

- 5) Noise is unavoidable:

Thermal noise: random electron motion

Modeled as AWGN (Additive White Gaussian Noise)

Signal quality:

- SNR = Signal / Noise
- In dB: $\text{SNR(dB)} = P_s - P_n$
- 6) We convert bits to signals (representation)

$$x(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT)$$

where: a_k = symbol values (from data bits) $g(t)$ = pulse shape (determines bandwidth)
 T = symbol period (symbol rate = $1/T$).

- 7) Optimal Receiver (Matched Filter) : Use a filter matched to the signal it Maximizes SNR at sampling time and Gives best possible detection.

Fundamental Tradeoffs :

- 1) Power vs Data Rate
- 2) Bandwidth vs Data Rate
- 3) Error Rate vs SNR
- 4) Efficiency vs Reliability

Wireless Communication

Why is wireless communication harder:

Unlike wired wireless communication has:

- Broadcast medium (everyone hears)
- Interference from others
- Signal weakens with distance
- Fading (unpredictable variations)

Therefore Wireless is fundamentally unreliable and shared

Wireless Link Budget : $P_r(\text{dBm}) = P_t(\text{dBm}) + G_t(\text{dBi}) + G_r(\text{dBi}) - L_{\text{path}}(\text{dB}) - L_{\text{misc}}(\text{dB})$

Margin calculation of : $\text{Margin} = P_r - S_{\text{min}}$

Fading:

Shadow Fading (Slow Fading): obstacles variation: $\sigma = 4 - 12$ dB

Multipath Fading (Fast Fading): Signals interfere, Variation: up to 30 dB

Modulation:

Quadrature Amplitude Modulation (QAM)

We transmit data using carrier waves:

$$s(t) = I \cos(2\pi fct) - Q \sin(2\pi fct)$$

- I and Q = independent signals
- Forms constellation diagrams

Adaptive Modulation

Modern systems adjust dynamically:

- Weak signal → low rate
- Strong signal → high rate

This is used in

- WiFi
- 4G/5G

Cellular: Frequency reuse

Divide area into cells and reuse frequencies far apart

Tradoff: More reuse → more capacity but more interference

Personal insight: Communication is fundamentally constrained by physics not our software. Everything depends on: Noise, Bandwidth (the limit of the speed) and Power (the reach) all of this explains things like why the wifi is slow as we move away.

Connection:

Physical layer constraints directly shape things like :

- 1) WiFi performance (adaptive rates)
- 2) Cellular design (cells, reuse)
- 3) Protocol choices (error correction, retransmission)

Question:

Which statement is true about transmission media?

- A. Twisted pair has higher bandwidth than optical fiber
- B. Optical fiber has lower attenuation than copper

- C. Coaxial cable supports longer distances than fiber
- D. All media have similar propagation speeds

Answer : B

Why is there no “best” transmission medium?

- A. Fiber is always superior in every scenario
- B. Each medium optimizes different tradeoffs (cost, distance, bandwidth)
- C. Media choice only depends on bandwidth

Answer: B

According to the Shannon-Hartley theorem:

- A. Capacity increases linearly with SNR
- B. Capacity increases logarithmically with bandwidth
- C. Capacity increases linearly with bandwidth
- D. Capacity is independent of SNR

Answer: C

Why do systems use adaptive modulation?

- A.To match data rate to channel quality
- B.To reduce bandwidth
- C.To eliminate noise
- D.To avoid interference completely

Answer: A

What is the benefit of frequency reuse?

- A.Reduces bandwidth usage
- B.Improves propagation speed
- C.Eliminates interference
- D.Increases total system capacity

Answer: D